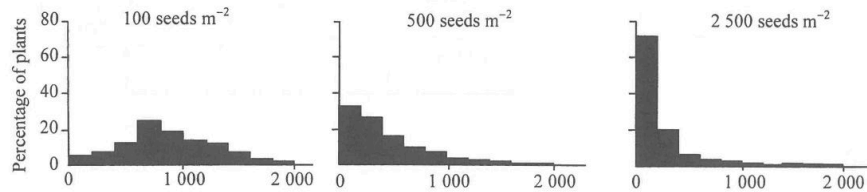


1. 2012Q13

Questions 13 and 14 refer to an investigation described below. Pots were planted with seeds at densities of 100, 500, and 2,500 m^{-2} respectively. The soil, water, and lighting conditions were similar for each pot. The dry masses of mature plants are shown in the following graphs.



The investigation was most likely performed to study

- A. the effect of symbiosis.
- B. the effect of competition.
- C. the success rate of seed germination.
- D. the optimum density for seed germination.

2. 2012Q14

Which of the following best describes the effect on the plants as the density of seeds planted increases?

- A. biomass of individual plants is reduced
- B. genetic variability of the plants is increased
- C. dry masses of plants are normally distributed
- D. leaves become smaller and stems get thinner

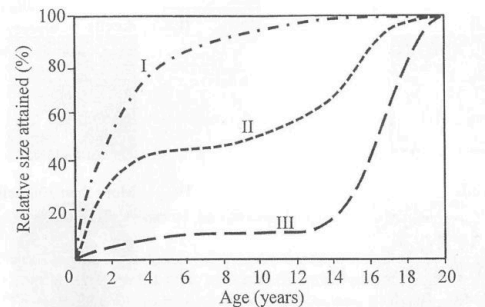
3. 2015Q28

Which of the following parameters is best for measuring the growth of the broad bean after germination?

- A. the length of the shoot
- B. the area of the leaves
- C. the weight of the embryo
- D. the volume of the cotyledon

4. 2017Q24

The graph below shows the growth curves of the head, reproductive system and whole body in humans:



Which of the following combinations correctly identifies curves I, II and III?

- | | I | II | III |
|----|---------------------|---------------------|---------------------|
| A. | whole body | reproductive system | head |
| B. | head | reproductive system | whole body |
| C. | head | whole body | reproductive system |
| D. | reproductive system | whole body | head |

5. 2018Q29

Which of the following substances contribute(s) most to the increase in biomass of a plant?

- A. water
- B. oxygen
- C. minerals
- D. carbon dioxide

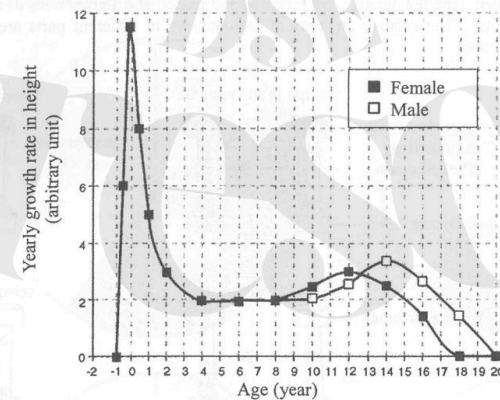
6. 2019Q30

Which of the following parameters is best used for recording the growth of a potted germinating seedling over a period of time?

- A. the dry mass of the seedling
- B. the fresh mass of the seedling
- C. the total surface area of the seedling's leaves
- D. the time taken for the seedling's first leaf to appear

7. 2019Q31

The graph below shows the growth rate of humans:



Which of the following can be deduced from the graph?

- A. The brain grows rapidly from age 0 to age 4.
- B. There is no more change in growth for males after age 20.
- C. The duration of adolescence is the same in males and females.
- D. Growth of the reproductive organs begins at age 10 in females.

8. 2020Q26

On which of the following parts of the root can root hairs be found?

- A. root cap
- B. region of elongation
- C. region of cell division
- D. region of differentiation

LQ

1. 2016Q9a

a) Briefly describe how the dry mass of a plant can be determined. (2 marks)

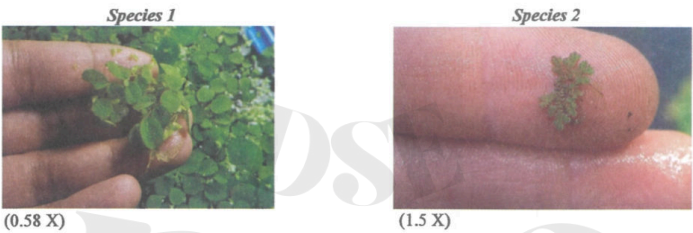
2. 2021Q5a

A student prepared cells of an onion root tip for observing cell division under a light microscope.

a) What type of cell division is likely to take place in the root tip of an onion? Explain your answer. (2 marks)

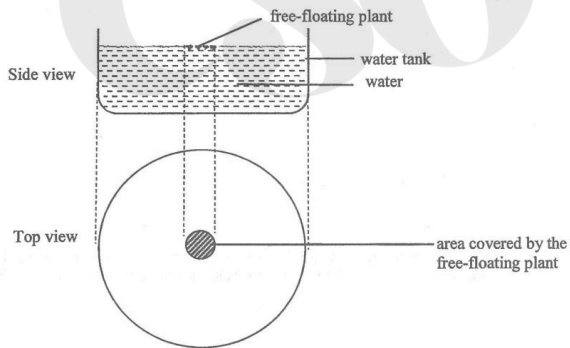
3. 2021Q9c

The photographs below show the appearances of two species of free-floating, freshwater plants, Species 1 and Species 2:

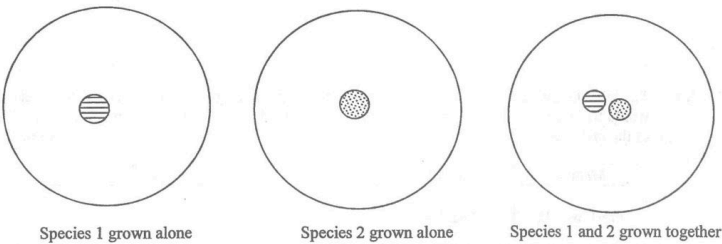


To study the interaction between these two plant species, each species was grown either alone or together with another species in a water tank for 50 days. Each species covered 10% of the area of water surface at the beginning of the experiment. The experimental set-up and design are shown in the following diagrams.

Experimental set-up:



Experimental design (top view):



c) The table below shows two other methods of measuring plant growth and whether these methods would be feasible in this experiment. Complete the table by giving justifications for the feasibility of the methods (2 marks)

Method	Feasibility	Justification
Fresh weight	Feasible	
Number of leaves	Not feasible	

Ans

2012Q13: B (54%) 2012Q14: A (61%) 2015Q28: A (59%) 2017Q24: C (87%)
 2018Q29: D (28%) 2019Q30: C (18%) 2019Q31: C (45%) 2020Q26: D (49%)

2016Q9a

- 9 (a) • dry the harvested plant in an oven at around 100°C (1)
 • until a constant mass is obtained upon repeated weighing (1) (2)

2021Q5a

- 5 (a) • mitotic cell division (1)
 • root tip is the vegetative part of the plant / is not the reproductive part of the plant / is the region for growth (1) (2)

2021Q9c

(c)

<i>Method</i>	<i>Feasibility</i>	<i>Reason</i>	
Fresh weight	Feasible	plenty of water supply, fresh weight will not be affected / the experiment can be conducted without killing the plants (1)	(2)
Number of leaves	Not feasible	leaf can grow in size without an increase in number / difficult to count / uncountable (1)	

DSE
 IESO